AZURE MEETUP PUGLIA

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2026

BENVENUTI

DevFinOps

The Convergence of Development, Finance and Operations in the Microsoft Azure Ecosystem

Carlo Sacchi

Global Azure Puglia 2026 – 17 aprile

Who I am

1. Senior DevOps & Cloud Engineer @ Metyis
   www.linkedin.com/in/carlo-sacchi

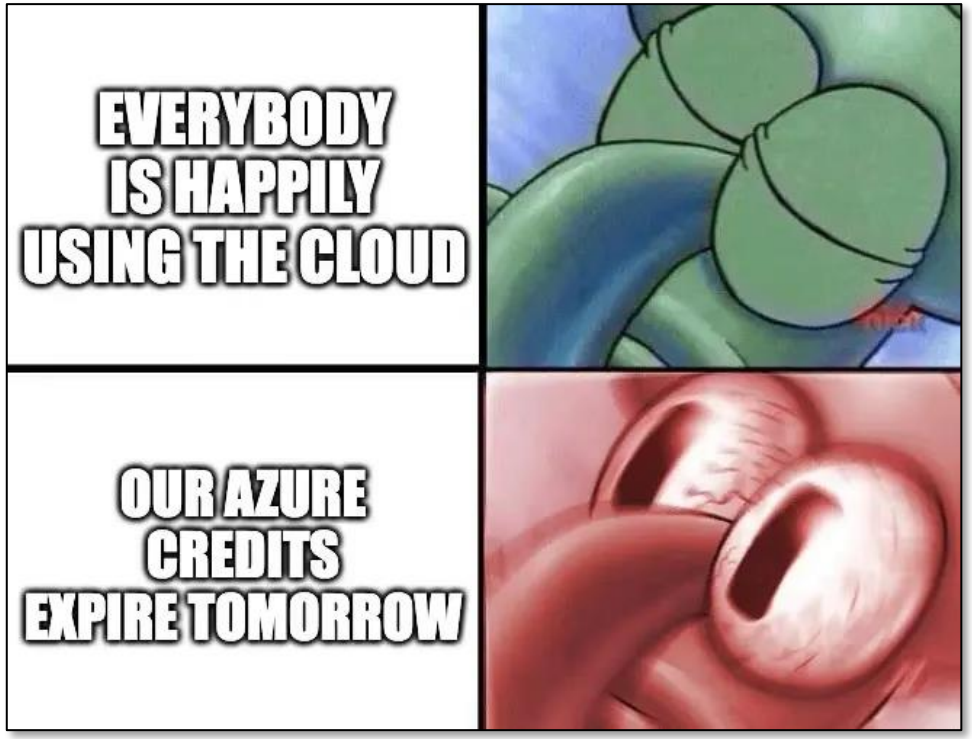
2. Azure Italia Podcast Host
   www.azureitalia.cloud

3. Runner and Outdoor passionate

What happens when IT teams ignore cloud costs?

Agenda

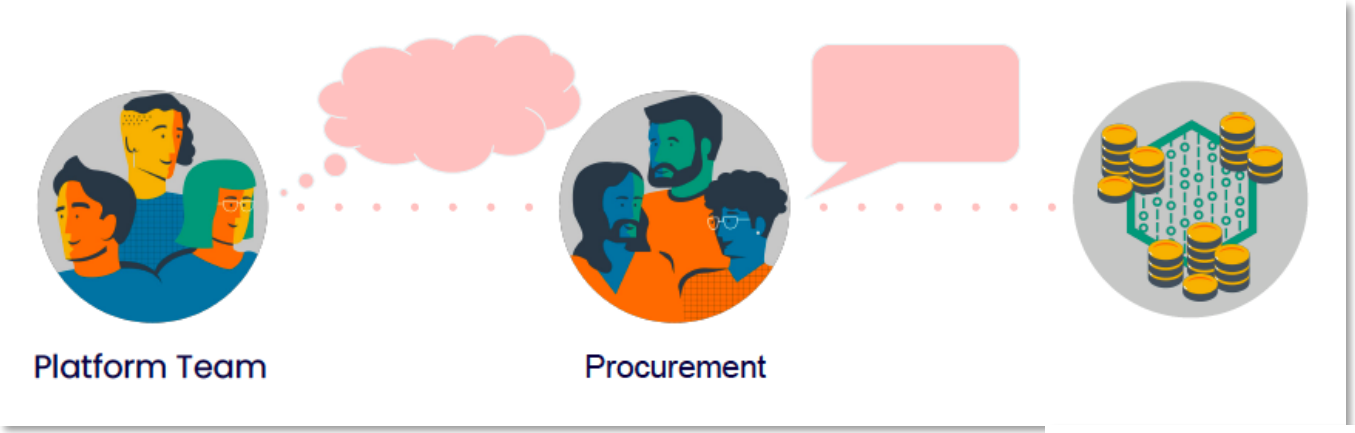
1. Why Cloud Costs Are Hard to Control
2. FinOps in a Nutshell
3. From DevOps to DevFinOps
4. DevFinOps in Microsoft Azure
5. Live Walkthrough: From Pipeline to Cost Insights
6. Key Takeaways


EVERYBODY
IS HAPPILY
USING THE CLOUD
OUR AZURE
CREDITS
EXPIRE TOMORROW

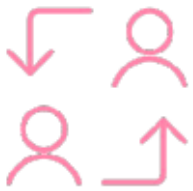
Devops teams to their finance team after misconfiguring their cloud computing needs
I am once again asking for your financial support.
yitz@komodor

Why Cloud Costs Are Hard to Control

Traditional Technology Consumption



Cloud Changes the Dynamic



Decentralized

buying centers for cloud are siloed from Finance and Central IT

Material

24% year over year increase in public cloud spend

Variable

Variable/consumption based cloud spend replacing datacenter/fixed cost spend

Inefficiency

Macroeconomic instability pressuring cloud spend and usage towards efficiency

FinOps in a Nutshell

and DevOps…

Understanding FinOps

- FinOps, short for (Cloud) Financial Operations, is an operational framework and cultural practice that aims:
  - to maximize the business value of cloud and technology (ROI)
  - Enables timely data-driven decision making
  - Creates financial accountability

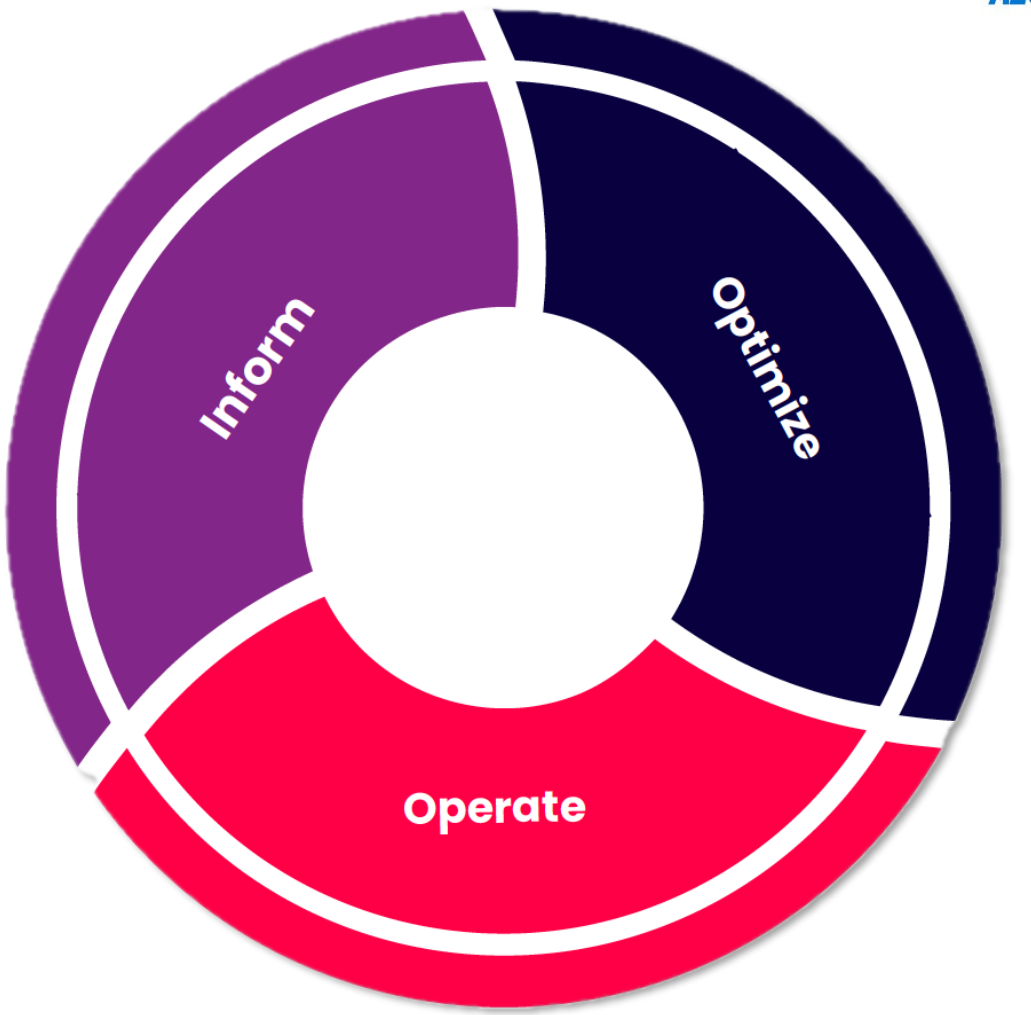
- Collaboration between engineering, finance, and business teams

The Three Phases of FinOps

1. **Inform**: Make cloud costs visible and well allocated

2. **Optimize**: Use that visibility to reduce waste and improve efficiency

3. **Operate**: Run and evolve workloads with cost in mind, in a continuous loop



FinOps Maturity Stages

- **Crawl**: Reactive, basic visibility and tagging
- **Walk**: Regular reviews, first optimizations and shared ownership
- **Run**: Cost is built into design, engineering and operations

Crawl

Walk

Run

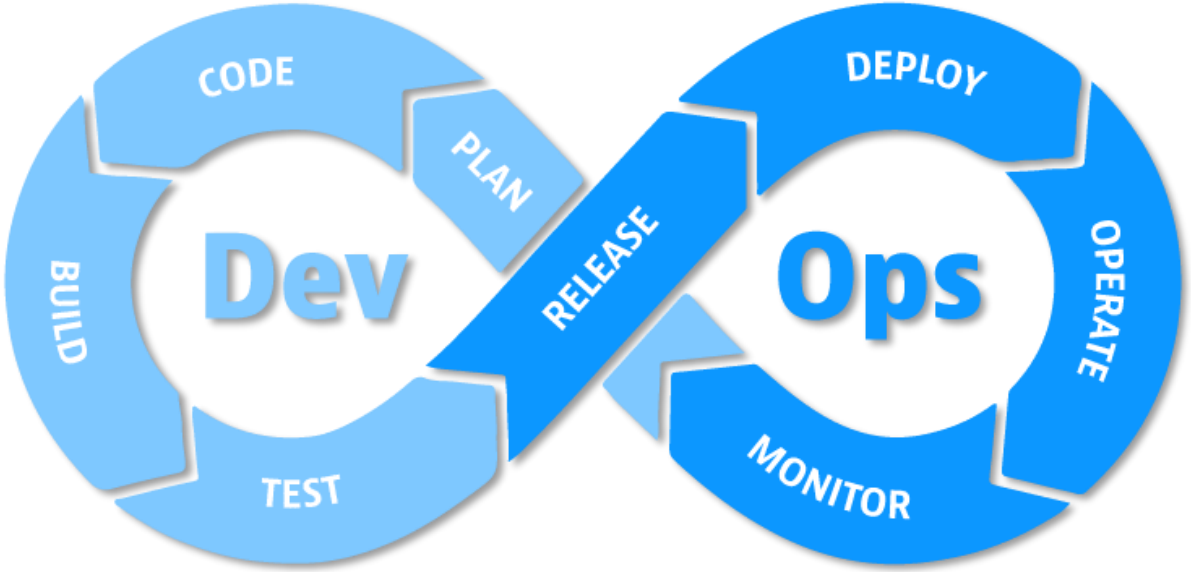
The 6 FinOps Principles

Developed by FinOps Foundation members, honed through experience.

- Teams collaborate, not work in silos
- Business value drives technology decisions
- Everyone owns their cloud usage and spend
- Cost and usage data is accessible, timely and accurate
- FinOps capabilities are enabled by a central platform
- Embrace the variable cost model of the cloud

Understanding DevOps

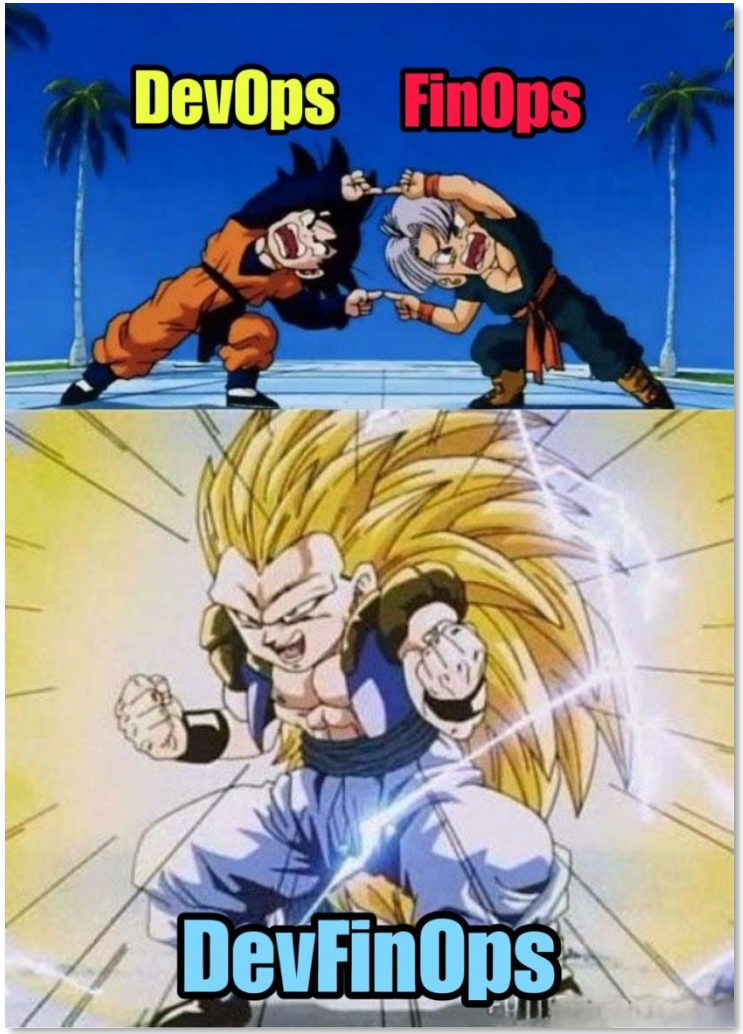
- Culture and practices that bring Dev and Ops together
- Shorter time from idea to production
- Continuous delivery
- Better, more reliable software



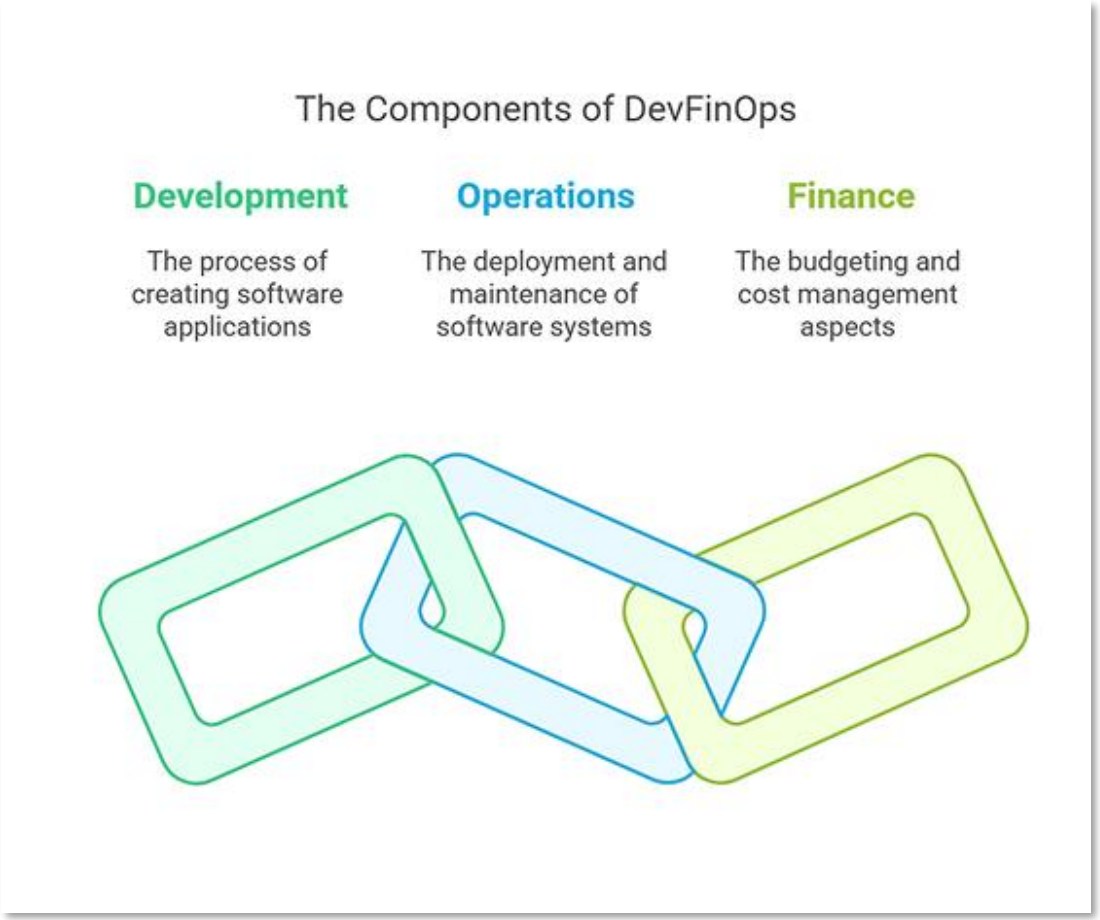
From DevOps to DevFinOps

Integrating FinOps and DevOps: The Concept of DevFinOps

- DevFinOps brings financial responsibility into the DevOps workflow.
- Cost is shared by Development, Operations and Finance, not owned by Finance alone.
- The result is a unified way of working where cost is a first-class concern, especially in the cloud.

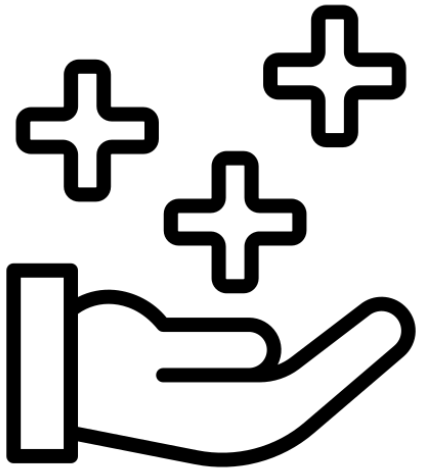


DevFinOps


The Components of DevFinOps
Development
The process of creating software applications
Operations
The deployment and maintenance of software systems
Finance
The budgeting and cost management aspects

Key benefits

- Better cost visibility and real-time spend insights
- Stronger collaboration between Dev, Ops and Finance
- Faster time-to-market without losing quality
- Higher efficiency and cost savings over time
- Decisions driven by data, not guesses



DevFinOps in Azure

Practical Implementation & Real-World Cases

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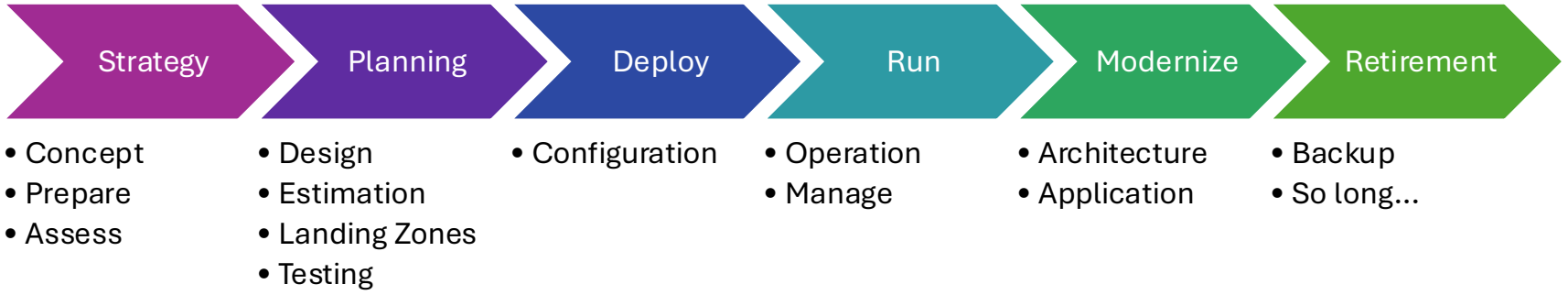
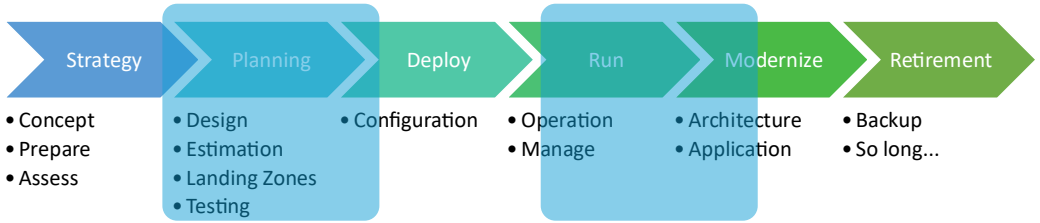
Cloud Adoption Journey & FinOps



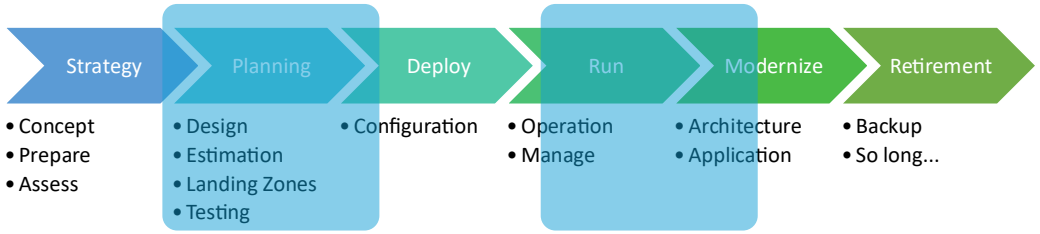
1) Strategic Resource Optimization in Azure

- Use **Reservations** and **Spot VMs** for rate optimization where workloads allow it.
- **Turn off idle resources** and **tune Storage**, **SQL** and **Backup** configurations.
- Leverage **Azure Migrate performance-based sizing** to plan migration costs.



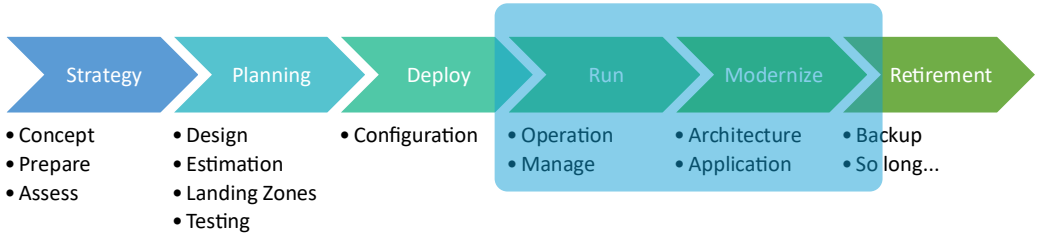
2) Governance with Azure Policy and Landing Zones

- Enforce **tagging**, **SKUs**, **locations** and cost-related limits centrally.
- Use Landing Zones as a cost-aware baseline for all subscriptions and apps.
- Support DevFinOps with **consistent guardrails** instead of manual checks.



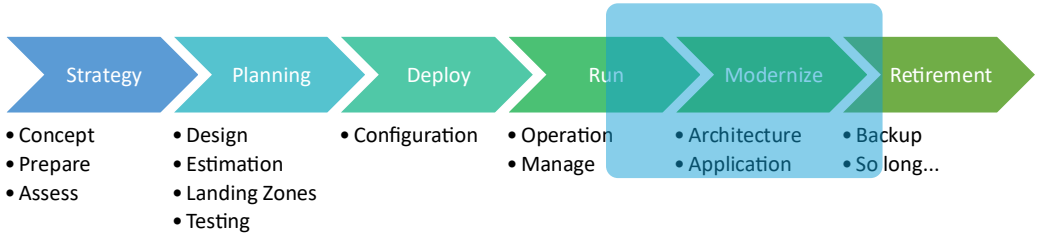
3) Azr Advisor

- Detect idle or under-used resources automatically.
- Get right-sizing and shutdown recommendations.
- Send regular digests or alerts to product and platform teams.
- Use Advisor reviews as a recurring DevFinOps check-up after deployment.

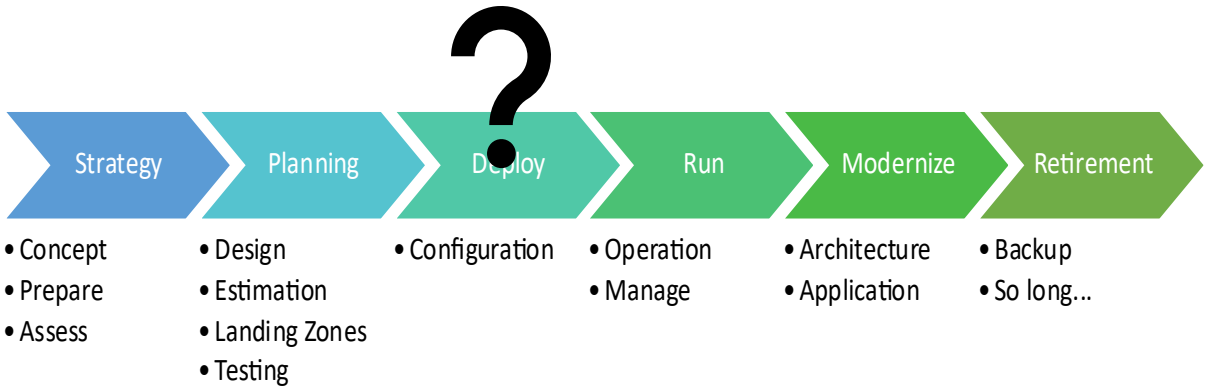


4) Monitoring with Azure NSG Flow Logs

- Gain visibility into real network traffic and patterns.
- Identify unused paths, noisy services and bandwidth-heavy workloads.
- Feed insights back into architecture and scaling decisions.
- Make network cost a regular part of the DevFinOps review cycle.



Deploy?


?
Strategy
Planning
Deploy
Run
Modernize
Retirement
• Concept
• Prepare
• Assess
• Design
• Estimation
• Landing Zones
• Testing
• Configuration
• Operation
• Manage
• Architecture
• Application
• Backup
• So long…

80

AUTOPILOT
START

I Found it! ACE!

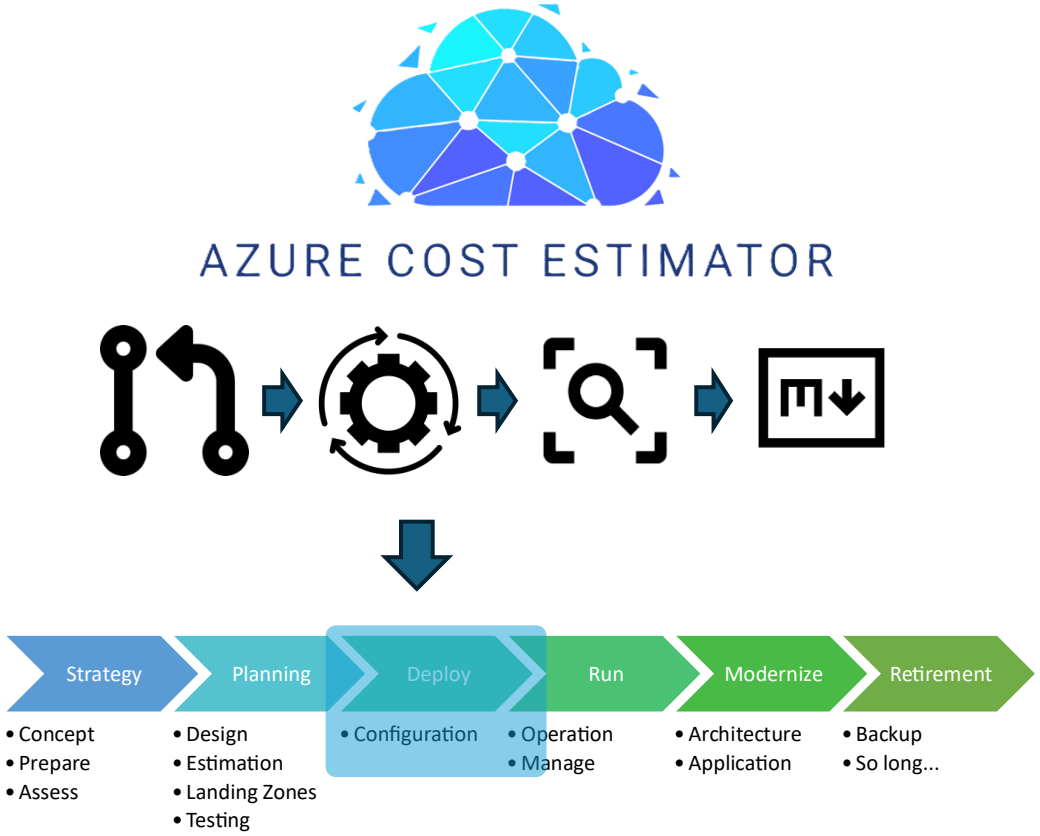
Azure Cost Estimator (ACE) is an open-source tool that turns your Bicep/ARM/Terraform templates into live Azure cost estimates directly in your CI/CD pipeline.

AZURE COST ESTIMATOR

Shifting FinOps Left with Cost Estimates in PR

https://github.com/TheCloudTheory/arm-estimator

- Use Azure Cost Estimator (ACE) in CI to **estimate** Bicep/ARM changes before deployment.
- **Show** expected monthly cost directly in the PR.
- **Fail** the PR if the forecast exceeds the agreed budget or policy thresholds.
- Make “how much will this cost?” part of every code review.



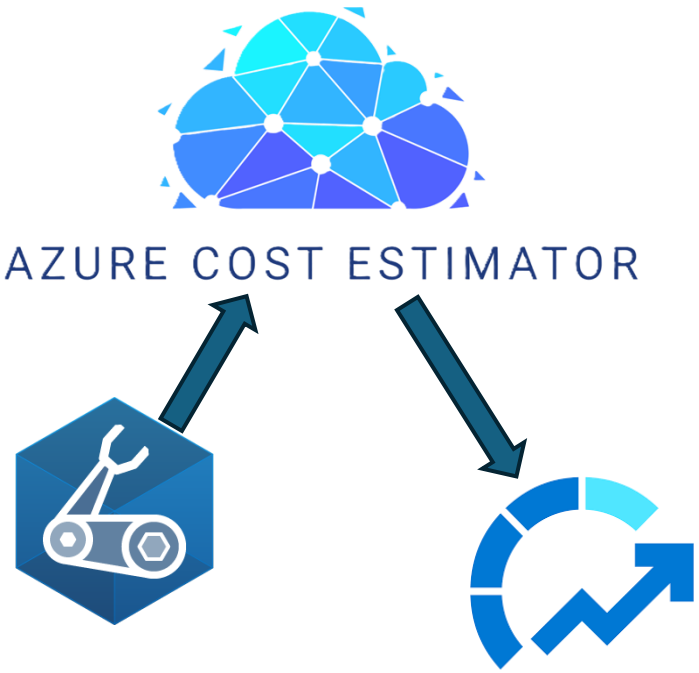
Some ACE Features

AZURE COST ESTIMATOR

- Free, Open-source (MIT license) project
- Docker or CLI
- Supports **49**+ Azure services (~**92** resource types)
- Bicep, ARM templates Native Support (Terraform early stage)
- Supports Incremental and Complete deployment modes
- Works with both standalone and nested resources
- 17 currencies supported
- Can export output as an artifact for further processing/reporting
- Supports custom usage patterns (hours, GB, transactions, etc.)
- Multiple scopes: RG, Sub, MGMTg, tenant

From Bicep/ARM/Terraform to concrete €/$

1. ACE parses Bicep, ARM, Terraform
2. For each resource, it extracts cost-relevant attributes (SKU, region, OS, storage tier, instance count, etc.)
3. Builds queries to the AZURE Retail Prices API https://prices.azure.com/api/retail/prices
4. Filters by service, SKU, region, currency, price type (Pay-As-You-Go)
5. Applies built-in or custom usage patterns
6. Aggregates everything into a per-resource and total monthly cost
7. Every run = fresh prices from Microsoft's official source



Docker or CLI

Docker run ...

```
docker run --rm \
  -e AZURE_CLIENT_ID="$AZURE_CLIENT_ID" \
  -e AZURE_TENANT_ID="$AZURE_TENANT_ID" \
  -e AZURE_CLIENT_SECRET="$AZURE_CLIENT_SECRET" \
  -v $PWD/infrastructure:/app/templates \
  -v $PWD/cost-reports:/app/cost-reports \
  thecloudtheory/azure-cost-estimator:latest \
  templates/main-compiled.json \
  $AZURE_SUBSCRIPTION_ID \
  ${{ env.RESOURCE_GROUP }} \
  --parameters templates/temp-params-${{ env.ENVIRONMENT }}.json \
  2>&1 | tee cost-output.log
```

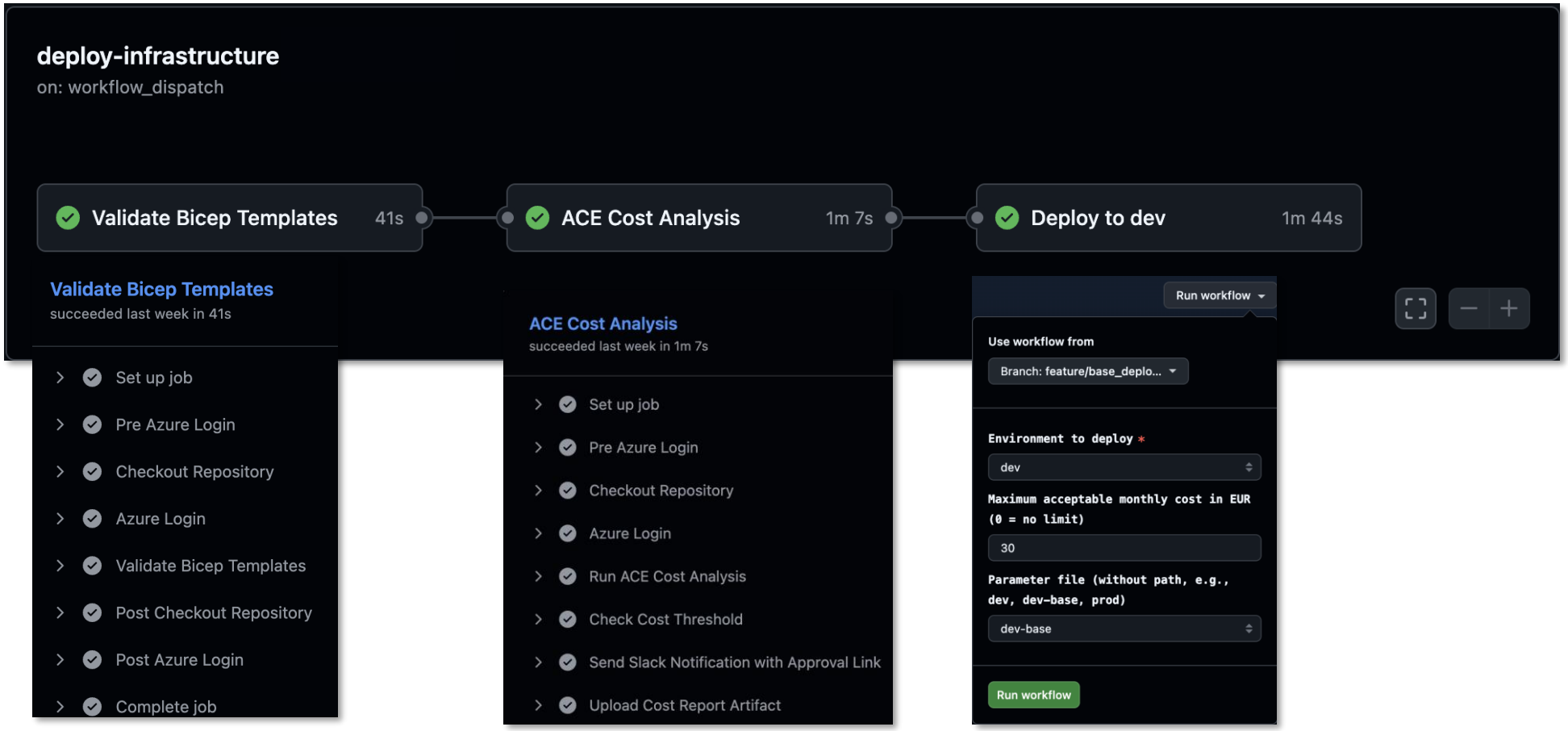
(install ACE CLI) then run command

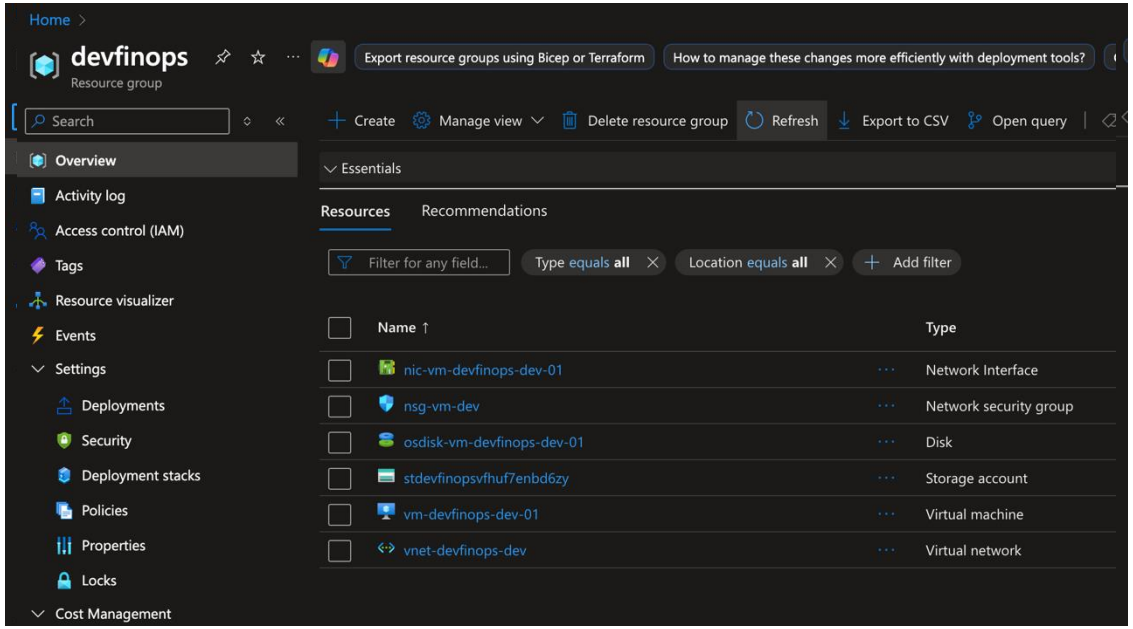
```
azure-cost-estimator \
  infrastructure/compiled/main.json \
  "${AZURE_SUBSCRIPTION_ID}" \
  "${RESOURCE_GROUP}" \
  --parameters infrastructure/compiled/params.json \
  --currency EUR \
  --generate-json-output \
  --json-output-filename cost-reports/cost-analysis.json \
  ${THRESHOLD_ARG} \
  2>&1 | tee cost-reports/ace-output.log
ACE_EXIT_CODE=$?
set -e
```

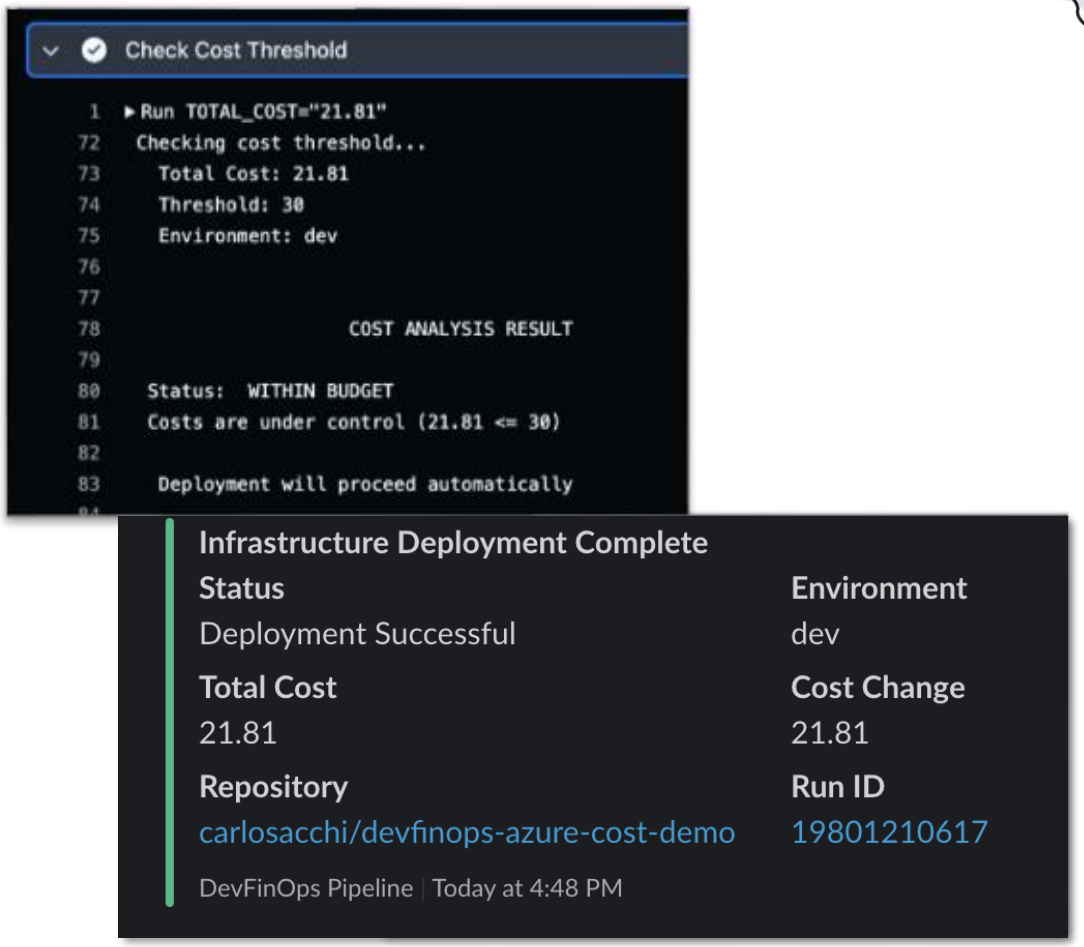
Live Walkthrough: DevFinOps Pipeline with ACE

1. Developer changes Bicep templates.
2. Pipeline runs ACE to estimate cost and compares it to the budget (30€).
3. Policy gate decides to block or proceed with deployment (1st attempt **ok**, 2nd **ko**).
4. Azure Budgets / alerts and monitoring track real spend after each release.

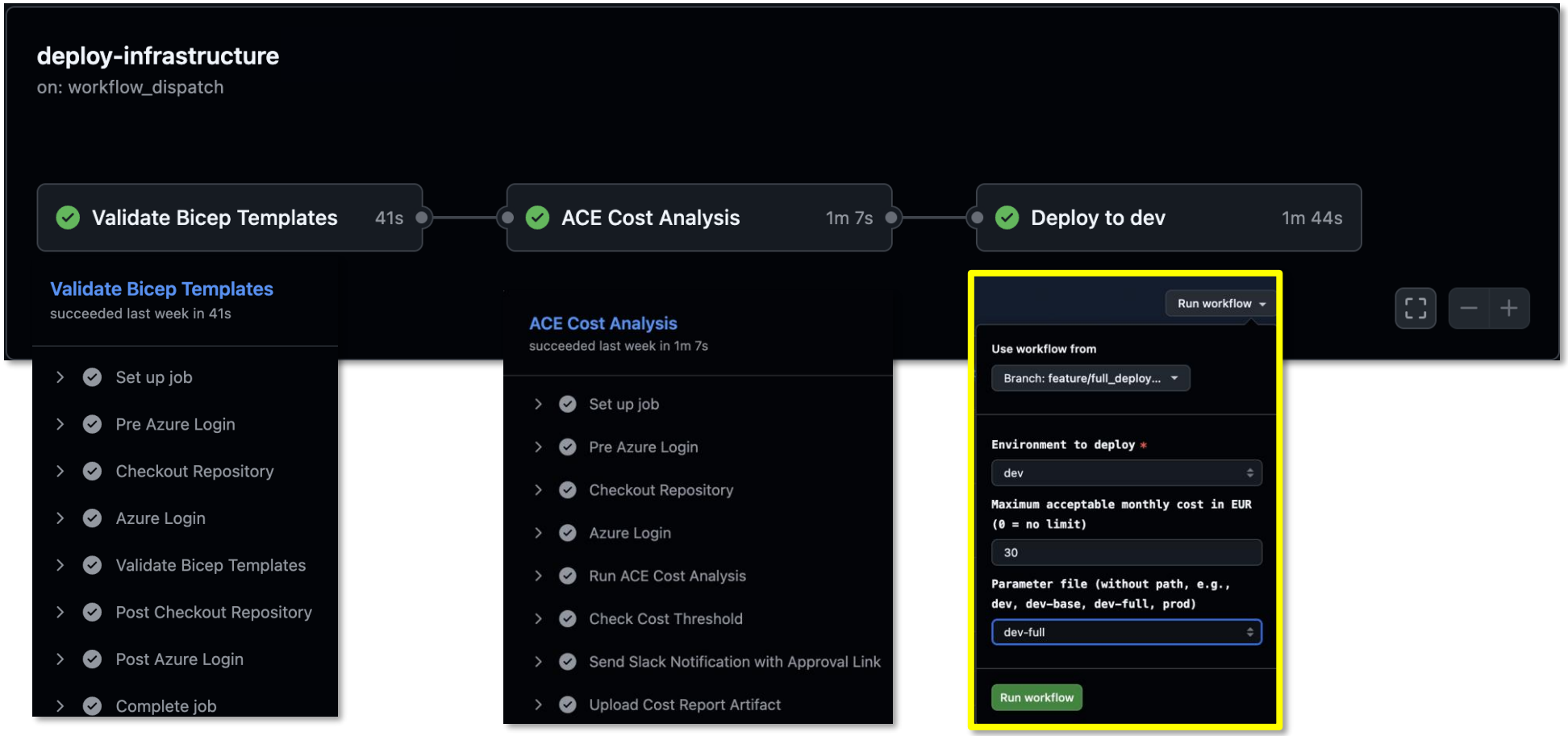
Live Walkthrough: DevFinOps Pipeline with ACE

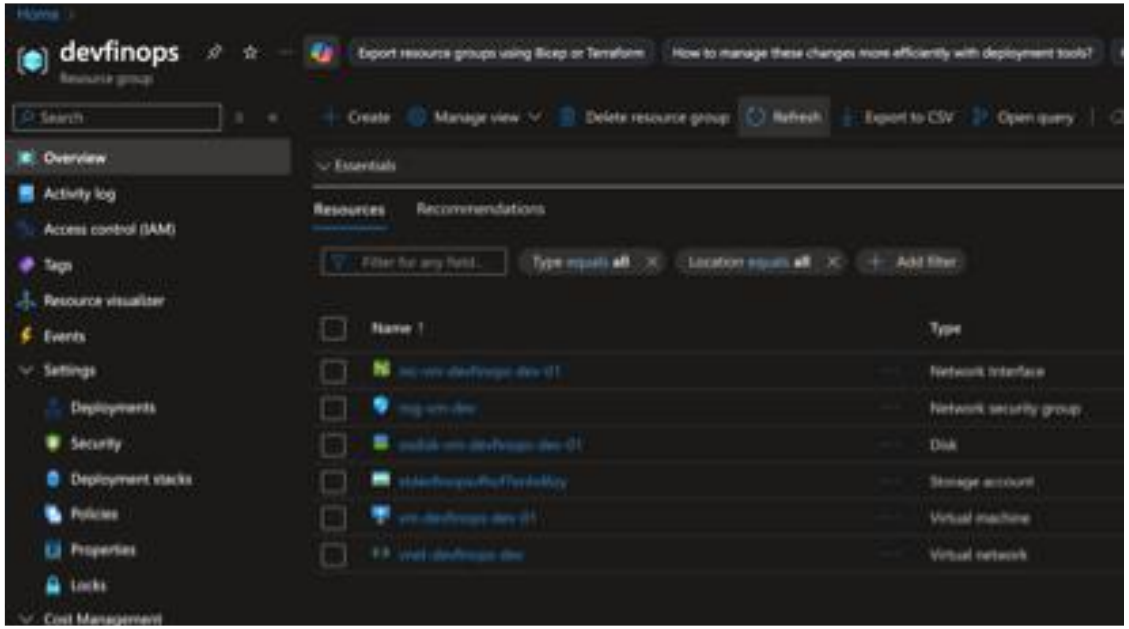



Home
devfinops
Resource group
Export resource groups using Bicep or Terraform
How to manage these changes more efficiently with deployment tools?
Search
Create
Manage view
Delete resource group
Refresh
Export to CSV
Open query
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Deployments
Security
Deployment stacks
Policies
Properties
Locks
Cost Management
Essentials
Resources
Recommendations
Filter for any field...
Type equals all
Location equals all
Add filter
Name
Type
nic-vm-devfinops-dev-01
Network Interface
nsg-vm-dev
Network security group
osdisk-vm-devfinops-dev-01
Disk
stdevfinopsvfhuf7enbd6zy
Storage account
vm-devfinops-dev-01
Virtual machine
vnet-devfinops-dev
Virtual network


Check Cost Threshold
Run TOTAL_COST="21.81"
Checking cost threshold...
Total Cost: 21.81
Threshold: 30
Environment: dev
COST ANALYSIS RESULT
Status: WITHIN BUDGET
Costs are under control (21.81 <= 30)
Deployment will proceed automatically
Infrastructure Deployment Complete
Status
Deployment Successful
Environment
dev
Total Cost
21.81
Cost Change
21.81
Repository
carlosacchi/devfinops-azure-cost-demo
Run ID
19801210617
DevFinOps Pipeline | Today at 4:48 PM

Live Walkthrough: DevFinOps Pipeline with ACE





Check Cost Threshold

```
1  ▶ Run TOTAL_COST="48.32"
72  Checking cost threshold...
73    Total Cost: 48.32
74    Threshold: 30
75    Environment: dev
76
77
78                    COST ALERT
79
80   Status:  OVER BUDGET
81   Costs exceed threshold (48.32 > 30)
82
83    Manual approval REQUIRED - costs exceed threshold!
84      Environment protection will require manual approval
```

ACE Cost Analysis 52s

5:09 PM **DevFinOps** APP Cost Analysis Complete

Deployment Cost Estimation - dev

Environment	Total Cost
dev	48.32/month
Cost Change	**Threshold**
26.51	30

Status
OVER BUDGET - Deployment BLOCKED

Cost Analysis
Costs exceed threshold (48.32 > 30)

Action Required
"None - proceeding automatically"

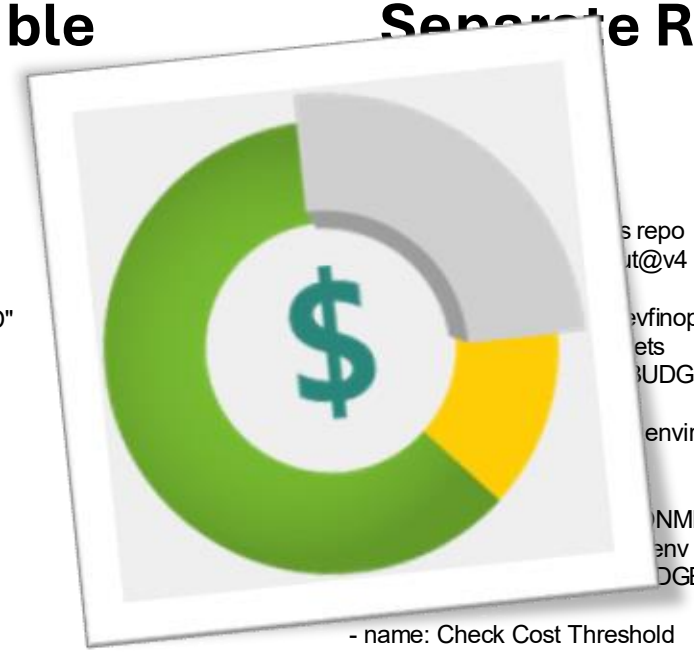
DevFinOps Pipeline · Today at 5:09 PM

Demo

How store budget value

GitHub Environment variable

```
env:
  COST_BUDGET_EUR: ${{ vars.COST_BUDGET_EUR }}
      - name: Check Cost Threshold
        run: |
          TOTAL_COST="${{ steps.ace-analysis.outputs.total-cost }}"
          THRESHOLD="${{ env.COST_BUDGET_EUR }}"
          echo "Total Cost: $TOTAL_COST"
          echo "Threshold: $THRESHOLD"
          EXCEEDS=$(awk -v cost="$TOTAL_COST" -v thresh="$THRESHOLD"
\
            'BEGIN { print (cost > thresh) ? "yes" : "no" }')
          if [ "$EXCEEDS" = "yes" ]; then
            echo "FAIL: Costs exceed threshold ($TOTAL_COST >
$THRESHOLD)"
            exit 1
          else
            echo "OK: Costs within threshold"
          fi
```



Separate Repos

```
s repo
ut@v4

evfinops-budgets
ets
BUDGETS_DEPLOY_KEY }}

environment


NMENT }}"
env "$ENV" '.[$env]' devfinops-budgets/budgets/app-xyz.json)
DGET" >> $GITHUB_OUTPUT

- name: Check Cost Threshold
    run: |
      TOTAL_COST="${{ steps.ace-analysis.outputs.total-cost }}"
      THRESHOLD="${{ steps.load-budget.outputs.budget-eur }}"
```

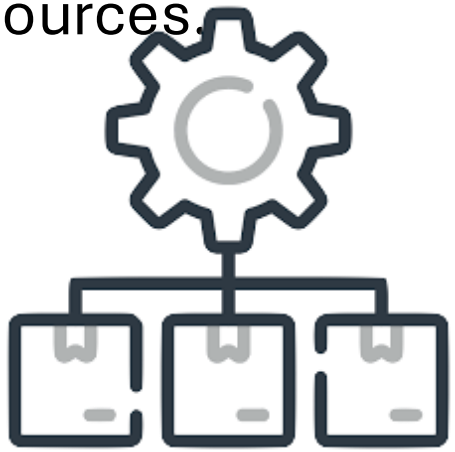
Key Takeaways

DevFinOps Lessons from the Field

- **Teams own the cost of the services they build**, not just the code.
- **Platform teams provide guardrails** and self-service tools, not manual approvals.
- **Unit economics (cost per feature / user / environment)** matter more than the total bill.
- **Cost becomes a product concern**, alongside reliability, performance and security.

Code Everything

FinOps-as-Code: Treat budgets, policies, and alerts as code, versioned and deployed via CI/CD, not just tagging resources.



Measure Everything Continuously

Cloud bills

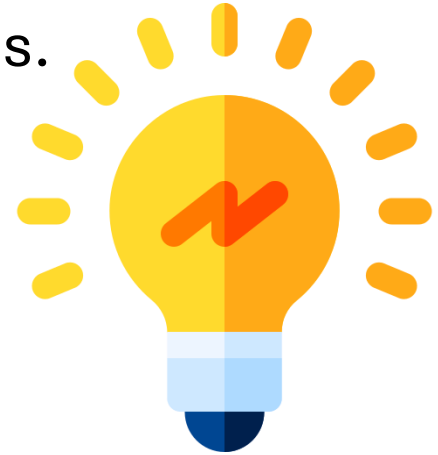
Application performance

Infrastructure resources

Deployments

What if cost per feature became a release KPI?

Bringing cost visibility into PRs and DORA metrics.



In a Nutshell

- DevFinOps: Shifting from Cost as a Risk to Cost as a Feature
- Make cloud costs visible early, not only at the end of the month.
- Bring Finance, Development and Operations into the same workflow.
- Use Azure-native tools to automate cost awareness and optimization.
- Turn cost from a scary invoice into a product KPI.

In a Nutshell

Introducing DevFinOps: The Strategic Shift Left

- Definition: FinOps (Cloud Financial Operations)
- "DevFinOps" Paradigm

Links

- https://github.com/TheCloudTheory/arm-estimator
- https://azure-cost-estimator.thecloudtheory.com/
- https://learn.microsoft.com/en-us/rest/api/cost-management/retail-prices/azure-retail-prices
- https://dev.to/vaib/revolutionizing-cloud-cost-management-the-power-of-ai-in-finops-3e2d

Emerging Trends

- AI-Driven FinOps
- SustainabilityOps

Questions

Feedback

Links

SCAN ME

Thank You